Cowrie SSH Honeypot on Ubuntu (Docker) This project documents the deployment and troubleshooting of a Cowrie SSH honeypot running in Docker on an older Mac repurposed with Ubuntu. Cowrie is a medium-to-high interaction SSH and Telnet honeypot designed to log brute-force attempts and attacker shell activity.

Overview:

The goal of this project was to build a small security lab that could: Run Cowrie in a containerized setup on Ubuntu. Isolate the honeypot from primary devices by using a separate network segment or guest Wi-Fi when available. Capture login attempts, session activity, and JSON logs for later review. Validate the deployment with controlled SSH testing from another machine

Environment:

Componet	**Details**
Host Hardware	Older iMac repurposed as a home lab machine
Host OS	Ubuntu
Container runtime	Docker / Docker Compose
Honeypot	Cowrie SSH honeypot
Test client	A separate machine used for SSH validation
Primary port	2222/tcp for Cowrie

Architecture:

```
    Internet
       |
Guest/Lab network
       |
Ubuntu host (old Mac)
       |
Docker container
       |
Cowrie SSH honeypot
```

Steps 1: Deployment

Install Docker. Docker was installed on Ubuntu so Cowrie could run in a containerized lab environment rather than directly on the host.

```
sudo apt update
sudo apt install -y docker.io docker-compose-plugin
sudo systemctl enable docker
sudo systemctl start docker
```

2. Create the project structure:

A dedicated project folder was created for configuration and persistent data.

```
mkdir -p ~/cowrie-docker
cd ~/cowrie-docker
mkdir -p cowrie-etc cowrie-var
```

3. Configure Docker Compose:

Cowrie’s Docker image supports port mapping, environment variables, and mounted configuration/data directories.

```
services:
  cowrie:
    image: cowrie/cowrie:latest
    container_name: cowrie-honeypot
    restart: unless-stopped
    ports:
      - "2222:2222"
    environment:
      - COWRIE_SSH_ENABLED=yes
      - COWRIE_TELNET_ENABLED=no
    volumes:
      - ./cowrie-etc:/cowrie/cowrie-git/etc
      - ./cowrie-var:/cowrie/cowrie-git/va
```

4. Create the Cowrie configuration:

A minimal configuration enabled SSH and JSON logging for later review.

```
[ssh]
enabled = true
listen_endpoint = tcp:2222:interface=0.0.0.0
```

[telnet]
enabled = false

[output_jsonlog]
enabled = true
logfile = /cowrie/cowrie-git/var/log/cowrie/cowrie.json

[output_textlog]
enabled = true

5. Create test credentials:

Cowrie credential acceptance is controlled through userdb.txt.
root:x:pass
admin:x:admin
test:x:test
ubuntu:x:ubuntu

6. Start the honeypot:

docker compose up -d
docker compose ps
docker logs cowrie-honeypot --tail 100

Troubleshooting:

This project included several realistic deployment issues that had to be resolved before the honeypot worked correctly.

Permission issues on mounted volumes:

Cowrie initially failed to write runtime files, such as the UUID under which the mounted var/lib/cowrie/uuid, cowrie-var directory was not writable by the container process.

Temporary troubleshooting fix:

sudo chmod -R 777 cowrie-var

Missing runtime directories:
Cowrie expects initialized directories, including var/log/cowrie, var/lib/cowrie/downloads, var/lib/cowrie/tty, and var/run for logs, downloads, session recordings, and runtime state.

Directories were created manually:

mkdir -p cowrie-var/log/cowrie
mkdir -p cowrie-var/lib/cowrie/downloads
mkdir -p cowrie-var/lib/cowrie/tty
mkdir -p cowrie-var/run

Login hanging after the correct password:

A test login using root/pass initially hung because Cowrie could accept credentials but could not finish creating the session playback log under var/lib/cowrie/tty/ until the missing directories and permissions were fixed.

Validation:

After the directory and permission fixes, the honeypot successfully accepted an SSH connection and opened the fake shell environment. Cowrie stores logs under and session playback files under var/lib/cowrie/tty/.

Example test command:

ssh -p 2222 root@<HONEYPOT_IP>

Password:

pass

Useful validation commands:

docker logs cowrie-honeypot --tail 100
find cowrie-var -type f | sort

This project helped build experience with:

- Docker-based service deployment on Ubuntu.
- Linux file permissions and volume-mount troubleshooting in containers
- SSH honeypot behavior and credential simulation with Cowrie.
- Log collection and session recording using Cowrie’s cowrie.json and TTY playback files.
- Basic home lab design and safer network isolation for security testing.

References and Documentation Used

- [Cowrie GitHub Repository](https://github.com/cowrie/cowrie)

- [Cowrie Official Documentation](https://docs.cowrie.org)

- [Cowrie Docker Documentation](https://docs.cowrie.org/en/latest/docker/README.html)

- [Cowrie Installation Documentation](https://docs.cowrie.org/en/latest/INSTALL.html)

- [Docker Compose Plugin Install Guide](https://docs.docker.com/compose/install/linux/)

- [Cowrie Docker Hub Image](https://hub.docker.com/r/cowrie/cowrie)